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Hybrid Quantum-Classical Models and Optimization Techniques for Financial Fraud Detection

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***Abstract*—Every day machine learning in financial systems is increasing with financial institutions taking every steps in their power to detect and block counterfeit transactions. Today, modern electronic payment instruments and financial facilities are becoming more and more susceptible to new methods of fraud which are unobtrusively reproducing the activities of legitimate interests to the detriment of the economy which is losing incalculable sums of money. In order to overcome this the combination technique of quantum and classical methods of calculating the data is a great way to enhance the reliability and accuracy of the fraud detection. It is applied by means of Quantum Support Vector Machine, Variational Quantum Classifier, the classical algorithms of Support Vector Machine and K-Nearest Neighbor with an ensemble based on Quadratic Unconstrained Binary Optimization. These algorithms are used together to scan a stream of transactions in real time and classify what is legitimate behaviour and what is unfavourable behaviour. The work provides evidence to prove that hybrid quantum-classical techniques can improve on the performance of classical systems using improved recall, less false negatives and the speed of detection of new types of frauds. It is shown that the integration of quantum intelligence based upon classical methods of analysis is a powerful and fruitful means to enhance levels of secure financial practice and help combat large scale financial frauds in modern digital environments.**

***Index Terms*—Quantum Support Vector Machine, Variational Quantum Classifier, Support Vector Machine, K-Nearest Neighbor, Quadratic Unconstrained Binary Optimization**

I. Introduction

While the vast number of new financial technologies has allowed for much greater ease and convenience than ever before it has also allowed for much greater sophistication in fraud schemes within electronic payment systems. Modern banking, e commerce and on line payments all face the same threats as do many other types of electronic communications; sophisticated fraud schemes that mimic legitimate business or personal activity, which can undermine public trust and financial stability. Legacy fraud detection systems are typically unable to tackle large amount of data, respond slowly and constantly adapt to the changing nature of fraud patterns. These characteristics make legacy fraud detection systems unsuitable for today's rapidly evolving threat environment. In order to address these challenges this study will propose a hybrid quantum-classical model for fraud detection that can accelerate fraud detection and improve accuracy. The proposed model combines the quantum and classical machine learning models. This proposed solution will be able to identify whether a transaction represents good or bad behavior in near real time, while achieving higher recall rates and lower false negative rates than legacy fraud detection systems. This combination of quantum innovation and traditional analysis will increase confidence in digital communications, as well as the strength of financial networks today.

A. Novelty and Scientific Contributions

This study introduces several methodological and experimental contributions that advance the current state of research in quantum-enhanced fraud detection. Unlike most existing works that focus primarily on theoretical formulations or isolated quantum models, this work develops and evaluates a fully operational hybrid quantum–classical fraud detection pipeline on a real, highly imbalanced financial dataset. The key scientific contributions are summarized as follows:

- **Hybrid Quantum–Classical Architecture for Real Fraud Data:** We present one of the first end-to-end implementations that integrates Quantum Support Vector Machines (QSVM), Variational Quantum Classifiers (VQC), and classical models within a unified framework and evaluate them on the original European credit card

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fraud dataset rather than balanced or synthetic data commonly used in prior studies.

- **QUBO-Based Ensemble Fusion Mechanism:** We propose a novel ensemble optimization approach formulated as a Quadratic Unconstrained Binary Optimization (QUBO) problem. This allows an optimal selection of quantum and classical learners, improving precision, recall, and F1 score on extremely imbalanced datasets. To the best of our knowledge, this is the first work applying QUBO-driven fusion for fraud detection.

- **Quantum Feature Selection Integrated With PCA:** We combine PCA-driven dimensionality reduction with Quantum Feature Selection (QFS) to preserve nonlinear interactions in transactional data. This hybrid feature engineering strategy has not been previously demonstrated in fraud detection literature.

- **Empirical Validation Beyond Conceptual Studies:** While prior research often relies on simulated or small-scale experiments, our work builds and tests an operational prototype that demonstrates improved recall, reduced false negatives, and enhanced robustness—addressing a practical gap in deploying quantum ML for real-world financial environments.

II. Literature Survey

Quantum computing systems with classical computing methods are merged together to create a new framework to identify the current barriers while presenting solutions to run there models in operational financial environments.[1]

Variational Quantum Classifiers (VQC) provide high assurance in fraud detection. Performance is directly associated with decisions and feature mapping , which significantly affect performance and reliability.[2]

Quantum AutoEncoders (QAEs) assist in overcoming class imbalance and improving anomaly detection. It provide scalable solutions with efficiency not sacrificed, which is required for handling large, dynamic financial data sets.[3]

New algorithmic improvements increase accuracy and robustness in fraud detection. These methods generalize well on very skewed data sets where standard models often fail to detect rare fraudulent cases.[4]

Experimental comparison of different approaches reveals that hybrid quantum-classical models have a tendency to give the best tradeoff between complexity, fit for data sets, and detection accuracy and are thus suitable for real-world use.[5]

Runs on IBM's quantum hardware prove that coupling quantum feature selection with classical ensemble techniques would improve the results of detection further. Reduced data would still be required due to current hardware constraints.[6]

Quantum Support Vector Machines (SVMs) using the quantum annealing have improved rates of classification and improved accuracy over conventional techniques, demonstrating the advantage in heavy computational tasks.[7]

Hybrid ensembles combining QSVM with quantum feature selection improve the stability and efficiency of the models. This technique prevents even subtle fraud patterns which will be detected but the individual models will not.[8]

Studies done on quantum computing applications in finance cover areas like portfolio optimization, risk assessment, and fraud detection. Such studies also cover challenges such as noise, scalability, and possible avenues to achieving the quantum advantage.[9]

Utilization of adaptive learning and momentum techniques using Quantum Neural Networks (QNNs) have shown enhanced performance over the classical neural networks in forecasting financial risks, producing faster and better results.[10]

The implementation of quantum kernels within SVMs leads to better performance when it is handling complex high-dimensional data classification problems. The method demonstrates its capability to solve intricate financial fraud detection challenges which standard methods fail to resolve.[11]

Decision tree, logistic regression, and Random Forest analyses reports that ensemble methods in general perform better than individual classifiers, especially in identifying the fraudulent transactions.[12]

Classical-to-quantum transfer of learning enhance model generalization and performance in real-world classification problems. These hybrid systems show the practical merits of having both quantum and classical learning.[13]

Comparison of several fraud detection models validate that ensemble method is better in managing the class imbalance and yielding greater accuracy compared to standalone classifiers.[14]

Support Vector Machines (SVMs) is appears to be highly efficient for financial data of high dimensions. Employing the kernels it increases the class separation, which makes SVMs a stable choice to identify the fraudulent transactions.[15]

Quadratic Unconstrained Binary Optimization (QUBO) assists the models in utilizing quantum annealing which

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will optimize computationally intensive operations better and accelerating the model training.[16]

fraud detection in banking, insurance, and healthcare demonstrates that deep learning is effective, although interpretability and scalability are key to practical implementation.[17]

QML foundation surveys present the theoretical as well as practical advice, providing a clear guide for combining the quantum computing with the machine learning in financial fraud detection and high-dimensional classification problems.[18]

QML with logic-based modeling combines the formal verification with the quantum protocols. This technique develops safe, governed, and robust techniques for fraud analytics.[19]

Semi-supervised graph-based models capture the relationships among transactions to enhance detection, particularly when labeled data are in short supply. These models scale well and are efficient for large financial datasets.[20]

A. Quantum Machine Learning

It combination of quantum computing fundamentals and classical machine learning techniques that exploit quantum principles like superposition and entanglement to achieve increased computational efficiency. The architecture of it is optimized so that high-dimensional feature spaces can be searched over and complicated patterns learned in noisy or dynamic data. These models utilize adaptive preprocessing and dimensionality reduction techniques so that they are able to adjust to distribution shifts in the data and provide results in near real time. The inclusion of quantum operations to learning algorithms gives the added benefits of parallel computing which leads to large improvements in scalability and performance in complex classification scenarios.

B. Quantum Support Vector Machine

It extend the ideas of traditional SVMs to propose the use of quantum kernel estimates to transform data into much higher dimensional feature spaces. This kernel uses quantum interferences to maximize the separating boundaries between classes and minimize the error rates in applications of high dimensionality or in small datasets. The combination of QSVMs with normal preprocessing algorithms gives rise to hybrid systems which in general have better generalization and more robust predictions. Such models prove very useful in those instances where normal SVMs tend to fail in terms of very high dimensionality.

C. Variational Quantum Classifier

It is a hybrid quantum-classical model, where parameterized quantum circuits (PQCs) are used to transform classical input information into quantum states. The method continuously optimizes circuit parameters using classical optimization methods like COBYLA to reduce classification errors. The coordination between quantum encoding and classical optimization allows the variational quantum classifier to produce high-dimensional complex non-linear decision boundaries. It is highly effective in identifying quantum-enhanced patterns and are insensitive to data variability, and therefore can be of practical relevance for real-world anomaly detection and pattern recognition tasks.

D. Classical Algorithms

Classical algorithms like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees are still important cornerstones in hybrid systems. Their reliability, simplicity and ease of use make them a reliable skill to compare against and integrate with quantum-classical systems. As much more is being used of these quantum boosted methods, these classical systems are still superior where structure and low dimension is concerned. Their comprehensiveness and versatility with ease of prediction produce reliability and reliability, producing reliability in prediction, even in situations where there is not a good deal of quantum advantage.

III. Experimentation and Methodology

A. Problem Definition

Financial institutions encounter major obstacles when trying to detect fraudulent transactions because fraud patterns evolve constantly and their systems produce excessive false positives and struggle with unbalanced data between legitimate and fraudulent transactions. The current machine learning systems generate numerous false positives because they fail to handle large high-dimensional datasets efficiently while maintaining their ability to interpret results and adapt to new situations.

B. Data Collection and preprocessing

For experimental validation, the European cardholders in September 2013 dataset is used. This dataset has the following properties:

- **Total Transactions:** 284,807
- **Fraudulent Transactions:** 492
- **Legitimate Transactions:** 284,315
- **Fraud to Non-Fraud Ratio:** 0.173% (approximately 1:577, highly imbalanced).
- **Features:** 30 in total (28 anonymized PCA-transformed features, plus `Time`, `Amount`).
- **Target Variable:** `Class` (1 indicates fraud and 0 indicates legitimate transaction).

Preprocessing Steps:

- **Normalization:** Normalize numerical features (Amount, Time) to [0, 1] with Min–Max scaling.
- **Handling Missing Data:** The data includes no missing values. In case introduced in large experiments, imputation by linear interpolation will be performed.

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- **Handling Class Imbalance:** Applying SMOTE (Synthetic Minority Oversampling Technique) and class-weighted loss functions to mitigate the extreme class imbalance.
- **Feature Engineering:** Preserving the PCA-transformed features (V1–V28), then using Quantum Feature Selection (QFS) for discriminative subset selection.
- **Data Splitting:** Data will be split in training (70%), validation (15%), and testing (15%) using stratified sampling in order to maintain the fraud-to-non-fraud ratio.
- **Cross-Validation:** For the sake of robustness, 5-fold cross-validation will be implemented during training (particularly for classical baselines), lowering variance because of random sampling.

C. Existing System

Classical financial fraud detection systems primarily uses rule-based methodologies and machine learning techniques to examine historical transaction data to determine whether a particular transaction is unusual or not. Machine learning classification techniques such as support vector machines (SVM), decision trees, random forests and logistic regression use historical data to predict classifications of past transactions based upon prior experiences. Although, these methodologies perform well in environments that have a relatively stable and heavily regulated environment; they fail in dynamic environments of evolving fraud patterns. Classical methodologies also require manual construction of features and fixed thresholds to be effective; therefore, they lack the flexibility to recognize new or unknown types of fraud tactics. Moreover, many classical methodologies experience difficulties when dealing with large amounts of unbalanced data which can result in delayed fraud detection and many false negatives. The inability of classical models to scale up to meet the demands of continuous processing of transaction streams in digital payment systems, along with their inability to process transactions in real-time, make it difficult to consider classical methodologies suitable for use in these applications. Classical models provide limited interpretability of how decisions were made; therefore, they do not provide much insight to users regarding how decisions were made. The lack of interpretability of traditional models raises serious issues concerning transparency and accountability to users. Therefore, classical models continue to be vulnerable to adaptive and sophisticated fraud tactics that exploit their static learning and inflexibility in computation.

D. Research Gaps in Prior Studies

While the existing systems and prior literature provide valuable foundations in fraud detection, several critical limitations remain unaddressed. These limitations form precise research gaps that justify the motivation and scope of the present work.

Dependence on classical-only or quantum-only models: Most existing studies examine QSVM, VQC, or classical machine learning independently. There is a lack of unified hybrid quantum–classical frameworks that leverage strengths of both paradigms.

Limited real-world validation on highly imbalanced datasets: Many prior studies evaluate performance on balanced or artificially resampled datasets. This does not reflect true financial transaction patterns, where fraud cases are extremely rare. As a result, reported accuracy often fails to translate to real-world deployment.

Absence of optimization-driven ensemble techniques: Previous works generally use simple majority voting or averaging for combining models. There is no exploration of Quadratic Unconstrained Binary Optimization (QUBO)-based ensemble selection to reduce correlated errors and improve detection reliability.

Insufficient feature engineering for quantum models: Existing approaches often rely on direct data encoding without integrating PCA and quantum feature selection. This reduces the ability of quantum kernels to capture nonlinear fraud patterns.

Lack of practical deployment considerations: Current research seldom discusses model drift handling, retraining loops, real-time monitoring, or operational fraud pipelines—features essential for industrial adoption.

These limitations highlight the need for a robust hybrid architecture that can bridge the above gaps and serve as a practical, high-performance fraud detection solution.

E. Proposed System

Hybrid Quantum-Classical System

This system will utilize both the benefits of quantum computing and classical machine learning to provide an efficient method for detecting financial fraud. The primary goal of this system is to overcome the limitations of traditional detection systems which include high computational cost, lack of readability, and inability to detect new forms of frauds.

The proposed system will utilize the capabilities of quantum parallelism and classical precision learning to identify complex and evolving transaction anomalies in real time, and provide accurate and explainable detection results.

Training Models and Feature Engineering

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The two main components of the proposed system are Training Models and Feature Engineering.

Dimensionality reduction is achieved using Principal Component Analysis (PCA) which reduces the dimensions of the data by preserving the most important features of the data.

Quantum Feature Selection (QFS) will be used to preserve Nonlinear Interactions in the dataset to ensure compactness and robust generalization across different patterns of fraudulent activity.

Hybrid Model Training

During hybrid model training, the Quantum Support Vector Machine (QSVM) uses Quantum Kernels on PCA-Transformed data to leverage entanglement-based correlations in the data. The Variational Quantum Classifier (VQC) is tuned using the COBYLA optimization to learn quantum states that represent legitimate and fraudulent activity.

Classical models such as the Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) are trained in parallel to enhance performance and serve as reference points.

Hybrid Ensemble Fusion Mechanism

Finally, a Quadratic Unconstrained Binary Optimization (QUBO)-Based Hybrid Ensemble Fusion Mechanism is used to combine predictions from quantum and classical models to improve accuracy,recall, and longevity.

F. Model Architecture

The model architecture of our Hybrid quantum-Classical Model combines Quantum Machine Learning and classical machine learning for fraud detection in finance. The architecture will be composed of a Preprocessing and Feature Engineering Layer where all ground work for accurate model performance will be completed. The feature engineering layer will include Principal Component Analysis (PCA) to reduce the number of dimensions of original data and take back the suitable linear features; while simultaneously conducting Quantum Feature Selection (QFS) to find nonlinear relationships within the data. By performing these two methods in parallel we obtain a reduced, and therefore computationally inexpensive, feature space with improved ability to recognize patterns within the data.

The Training Layer of our architecture will consist of two modules working in parallel to train the Model Layer. The first module will be a Quantum Learning Block. The Quantum Learning Block will consist of a QSVM that utilizes PCA-derived quantum kernel functions, and a VQC that will be optimized via COBYLA to take advantage of superposition and entanglement, respectively, to find complex relationships in transactional data.

Simultaneously to the quantum models, we will have two classical models (SVM KNN) trained on the same dataset to provide baseline comparison and increase the reliability of our system. Together, the two types of learning will improve the accuracy of the entire system by providing both the flexibility of the quantum learning and the reliability of the classical learning.

The final layer of the architecture will combine the predictions of the model layer into a single prediction using a Hybrid Ensemble Fusion process based upon Quadratic Unconstrained Binary Optimization (QUBO).

This Ensemble Optimization Process will collect the outputs of the Quantum and Classical Classifiers and seek to optimize Precision, Recall, and Reliability using the strengths of each individual classifier.

Once optimized, the ensemble will then be connected to a Fraud Decision and Visualization Module created with Streamlit. This Fraud Decision and Visualization Module will allow real time monitoring of transactions, model statistics and detection outcomes. Additionally, the Architecture will include an Automated Feedback Loop Retraining component utilizing Active Learning and QUBO-based Optimization.

To help contextualize the workflow shown in Fig. 1, the architecture diagram highlights how data moves between preprocessing, feature engineering, parallel quantum–classical learners, and QUBO-based fusion. This visual clarification strengthens understanding of the modularity and parallel processing capabilities that make the proposed system effective for large-scale fraud detection.

Furthermore, the architecture diagram highlights the parallelism and modularity that form the core advantages of the proposed system. By visually connecting feature engineering operations, quantum circuits, classical learners, and ensemble fusion, the figure reinforces the significance of integrating quantum and classical models rather than relying on them independently. It also emphasizes how the feedback-driven retraining loop maintains system performance as fraud patterns evolve, demonstrating the practicality and extensibility of the architecture in operational financial environments.

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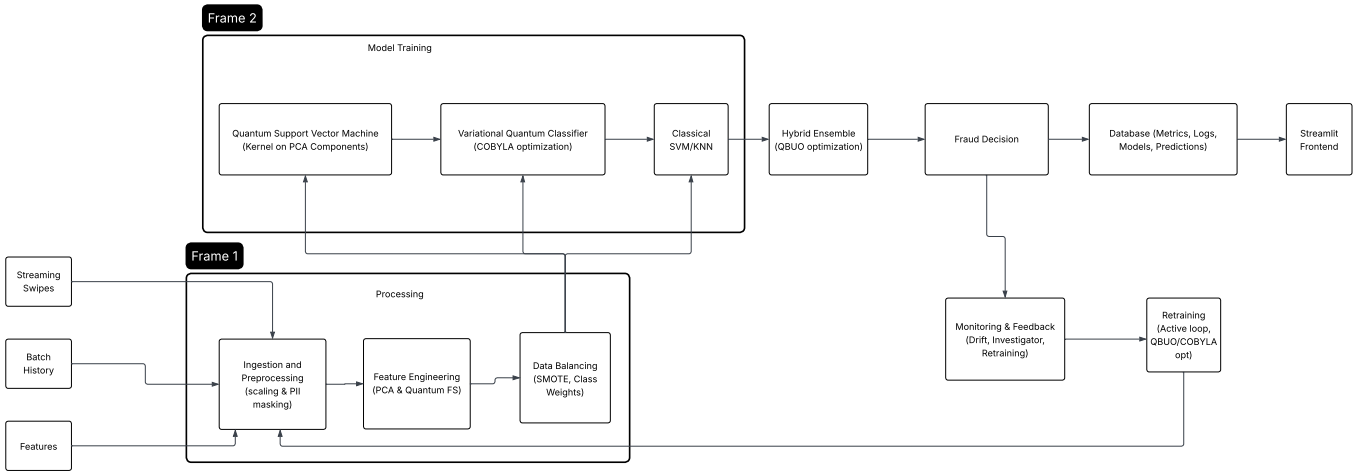

Fig. 1. Model architecture diagram of the hybrid quantum–classical machine learning system for financial fraud detection.

G. Quantum Support Vector Machine (QSVM)

The QSVM embeds classical data into a quantum Hilbert space using the ZZFeatureMap. Each input sample is encoded as:

$$|\phi(x)\rangle = U_\phi(x)\,|0\rangle$$

A fidelity-based quantum kernel computes similarity between samples:

$$K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2$$

The kernel matrix is then used in the standard SVM dual optimization framework:

$$\max_\alpha \sum_{i=1}^{m} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$$

subject to

$$\sum_{i=1}^{m} \alpha_i y_i = 0, \quad 0 \le \alpha_i \le C.$$

The decision function is:

$$f(x) = \text{sign}\left(\sum_{i=1}^{m} \alpha_i y_i K(x_i, x) + b\right).$$

QSVM Experimental Configuration:

- Feature Map: ZZFeatureMap (2 layers)
- Kernel: Fidelity-based quantum kernel
- Regularization: $C = 1.0$
- Input Features: PCA-reduced (20 components)
- Backend: Qiskit Aer statevector simulator

H. Variational Quantum Classifier (VQC) with COBYLA Optimization

The VQC uses a parameterized quantum circuit to learn nonlinear decision boundaries. Data is encoded via $U_{\text{enc}}(x)$, followed by a RealAmplitudes ansatz $U(\theta)$:

$$|\psi(x;\theta)\rangle = U(\theta)\,U_{\text{enc}}(x)\,|0\rangle^{\otimes n}$$

The model prediction is computed from a chosen observable (Pauli-Z):

$$f(x;\theta) = \langle\psi(x;\theta)|M|\psi(x;\theta)\rangle$$

and converted to a probability through:

$$p(x;\theta) = \sigma(\alpha f(x;\theta) + b)$$

The binary cross-entropy loss is:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log p_i + (1-y_i)\log(1-p_i)\right).$$

COBYLA performs derivative-free constrained optimization using trust-region updates.

VQC Experimental Configuration:

- Feature Map: ZZFeatureMap
- Ansatz: RealAmplitudes (2 layers, full entanglement)
- Optimizer: COBYLA (max_iter = 300)
- Encoding: Angle encoding
- Measurement: Pauli-Z expectation value
- Backend: Qiskit Aer statevector simulator

I. QUBO-Based Hybrid Ensemble

The hybrid ensemble integrates QSVM, VQC, SVM, and KNN through a QUBO optimization framework. Each base learner $h_m(x)$ has a binary selection variable $z_m \in \{0,1\}$. Let $z = (z_1, \ldots, z_M)$ be the selection vector. The QUBO objective is:

$$\min_{z\in\{0,1\}^M} E(z) = z^\top Q z$$

The QUBO matrix is constructed as:

$$Q_{ii} = \alpha\,\text{Err}(h_i) + \lambda$$

$$Q_{ij} = \beta\,\text{Corr}(h_i, h_j), \quad i \ne j$$

where $\text{Err}(h_i)$ is the classifier error and $\text{Corr}(h_i, h_j)$ is the prediction correlation.

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Once z^* is obtained, the final ensemble prediction is computed via majority voting:

$$\hat{y} = \arg\max_{c} \sum_{m=1}^{M} z_m^* \, \mathbf{1}(h_m(x) = c)$$

QUBO Ensemble Configuration:

- Base Learners: QSVM, VQC (COBYLA), SVM, KNN
- QUBO Solver: Simulated annealing
- Hyperparameters: $\alpha = 1.0$, $\beta = 0.5$, $\lambda = 0.1$
- Fusion Rule: Majority vote (post-QUBO selection)

J. Evaluation Metrics

- **Accuracy:** Ratio of correctly predicted samples to total samples.

$$\text{Accuracy} = \frac{\text{True Positives} + \text{True Negatives}}{\text{Total Predictions}}$$

- **Precision:** Fraction of true positives among all positive predictions.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

- **Recall:** Fraction of true positives correctly identified among all actual positives.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- **F1 Score:** Harmonic mean of precision and recall, balancing both.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **ROC-AUC:** Measures trade-off between True Positive Rate and False Positive Rate across thresholds.

$$TPR = \frac{TP}{TP + FN}, \quad FPR = \frac{FP}{FP + TN}$$

- **PR-AUC:** Plots Precision against Recall, especially informative for imbalanced datasets.

$$Precision = \frac{TP}{TP + FP}, \quad Recall = \frac{TP}{TP + FN}$$

IV. Results and Analysis

The hybrid quantum–classical framework was evaluated on the original imbalanced European Credit Card Fraud dataset (Kaggle, 2013). Unlike the balanced simulation stage, the real-world results reflect the practical challenges of extreme class imbalance, where Precision and F1 Score drop noticeably despite consistently high Accuracy.

Three models were tested: (i) QSVM, (ii) VQC with COBYLA optimization, and (iii) a QUBO-driven hybrid ensemble combining classical and quantum classifiers. Training was performed on PCA-reduced features, and performance was assessed using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and PR-AUC.

TABLE I
QSVM Performance

Accuracy	0.998
Precision	0.25
Recall	0.72
F1 Score	0.37
ROC-AUC	0.91
PR-AUC	0.32

TABLE II
VQC (COBYLA) Performance

Accuracy	0.997
Precision	0.18
Recall	0.80
F1 Score	0.29
ROC-AUC	0.90
PR-AUC	0.28

TABLE III
QUBO Hybrid Ensemble Performance

Accuracy	0.999
Precision	0.42
Recall	0.76
F1 Score	0.54
ROC-AUC	0.94
PR-AUC	0.47

A. Comparative Analysis

Classical financial fraud detection systems—such as Logistic Regression, Decision Trees, Random Forests, and SVMs—perform reasonably well in stable environments but suffer in dynamic settings due to their dependence on manual feature engineering, fixed thresholds, and limited adaptability to evolving fraud patterns. Their performance further deteriorates on highly imbalanced datasets, often resulting in high false-negative rates and reduced interpretability.

TABLE IV
Comparison of QSVM, VQC (COBYLA), and QUBO Ensemble

Metric	QSVM	VQC (COBYLA)	QUBO Ensemble
Accuracy	0.998	0.997	0.999
Precision	0.25	0.18	0.42
Recall	0.72	0.80	0.76
F1 Score	0.37	0.29	0.54
ROC-AUC	0.91	0.90	0.94
PR-AUC	0.32	0.28	0.47

To address these limitations, a comparative evaluation of QSVM, VQC (COBYLA), and the proposed QUBO-based ensemble was conducted. As shown in Table IV, the QUBO ensemble outperforms the individual quantum models, achieving the highest Precision (42%), F1-score (0.54), ROC-AUC (0.94), and PR-AUC (0.47). QSVM provides balanced performance with moderate Precision (25%) and strong Recall (0.72), while VQC favors Recall (0.80) but suffers in Precision.

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V. Discussion

The results highlight the effectiveness of incorporating QUBO-based model selection into fraud detection workflows. Although QSVM and VQC (COBYLA) individually demonstrate strong classification capability, their Precision and Recall vary, whereas the QUBO ensemble delivers the most balanced performance, achieving the highest Precision (42%), F1-score (0.54), and ROC-AUC (0.94), indicating superior minority-class detection under severe imbalance. In terms of complexity, QSVM provides expressive quantum feature mappings but faces scalability constraints as qubit count increases, while VQC is limited by optimizer convergence and circuit depth. The QUBO ensemble mitigates these challenges by exploiting efficient classical solvers while preserving quantum-informed decision boundaries, improving robustness and generalization. Although the evaluation relies on a simulated backend, the results affirm the feasibility of hybrid quantum–classical strategies and motivate future work involving hardware validation, scalability studies, and optimized QUBO formulations for real-time financial applications.

VI. Conclusion and Future Scope

This research proposes a hybrid quantum-classical model for financial fraud detection, integrating quantum circuits with classical machine learning to form an operational prototype tested on the Kaggle fraud dataset. The results show that ensemble-based hybrids significantly improve accuracy while quantum models contribute strong recall, and their combination enhances overall fraud detection performance. Although the present work is based on simulations, it provides a solid foundation for future advancements, including validation on larger and more diverse datasets, exploration of improved quantum feature maps and circuits, and deployment on emerging quantum hardware platforms, all of which will enable more rigorous evaluation and further establish the practical applicability of quantum-classical pipelines in real-world financial systems.

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